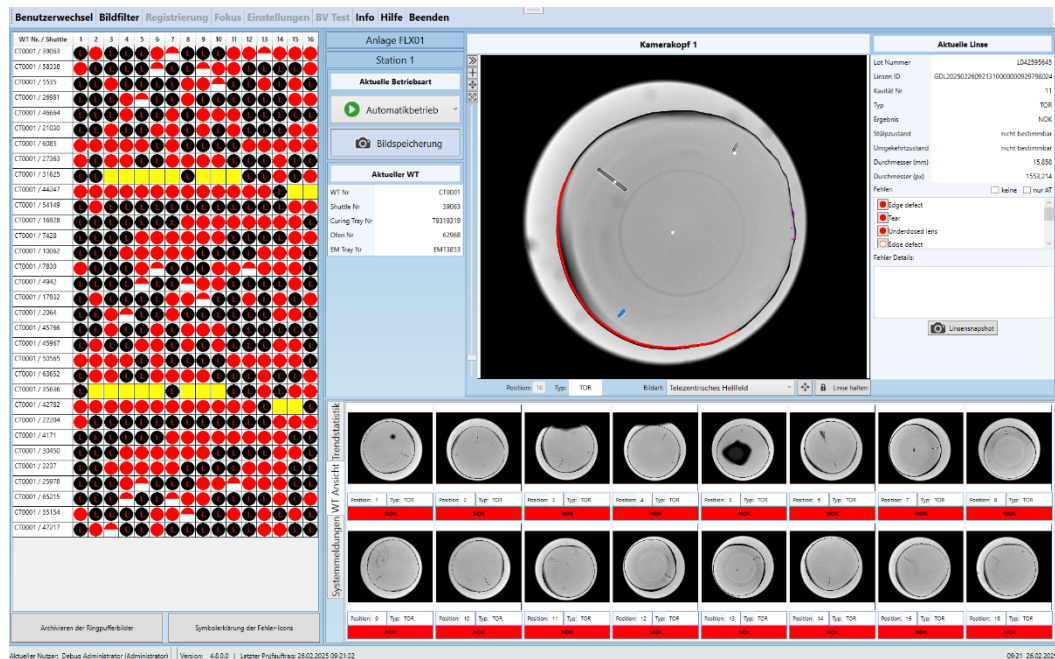




Manual OKLIN 3

Optical contact lens inspection OKLIN3

Version 5.x.y

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Document Name	Date	Amendments	Valid from Oklin3		
			3.x.y	4.x.y	5.x.y
Oklin3 Handbuch_V1.0.docx	01.07.2021	Initial version	3.12.0	4.1.0	---
Oklin3 Handbuch_V2.0.docx	14.07.2022	3.13.0: IDs 116, 118, 121, 132, 143 4.4.0: IDs 94, 117, 119, 122, 133, 144	3.13.0	4.4.0	---
Oklin3 Handbuch_V2.1.docx	26.09.2022	4.5.0: IDs 205, 178	---	4.5.0	---
Oklin3 Handbuch_V3.0.docx	03.05.2023	4.6.0: ID 17, 146, 268, 270, 271, 272, --- 275, 276, 277, 283		4.6.0	---
Oklin3 Handbuch_V3.1.docx	03.05.2024	Adjustment of text sections	---	4.6.0	---
Oklin3 Handbuch_V4.0.docx	03.06.2024	4.7.0: ID 335, 355	---	4.7.0	---
Oklin3 Handbuch_V5.0.docx	23.09.2024	4.8.0: ID 388, 413, 385	---	4.8.0	---
Oklin3 Handbuch_V6.0.docx	15.02.2025	4.9.0: ID 429, 430	---	4.9.0	---
Oklin3 Handbuch_V7.0.docx	03.06.2025	5.0.0: ID445, ID447, ID456	---	---	5.0.0
Oklin3 Handbuch_V8.0.docx	19.09.2025	5.1.0: ID499, ID502, screenshots updated	---	---	5.1.0

1 Introduction

1.1 General

This manual contains information for the operation of the Oklin3 software for inspection of contact lenses.

For different production facilities, the software can be specifically configured regarding certain properties. These include, for example, the number of camera heads, cameras and lenses. In addition, it is possible to set up the software for the inspection of existing contact lens images for the evaluation of the underlying image processing algorithms. This variant can be used offline and does not require any connection to additional hardware. The evaluation software is described in a separate document.

1.2 Service contact

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2 Installation

For the installation of the production software, the prerequisite is that the IT of the site has played the test computer with Horizon. If this requirement is not met, it is not advisable to continue the installation.

The configuration of the software for the production or evaluation software of the desired system type takes place via an installation program. A description of the complete installation of a new PC in the system, including hardware control and configuration of additional software, can be found in the document "Oklin3_V3-x-y_Desasterplan_V1.1.pdf" for Oklin3 3.x.y or "Oklin3_V4-x-y_Desasterplan_DE_V1.0.pdf" for Oklin3 4.x.y.

If Oklin3 is started from a folder other than C:\MA, a warning is displayed in the Oklin3 system message. However, the software continues without any problems.

2.1 Folder Structure

▼ Oklin3	Main folder
bin	Oklin3 module libraries
CsvStatistic	Location for statistics files
de	Language libraries
> doc	Manuals
Evaluation	Location of the error name configuration (Eval tool)
▼ Inbox	Inbox Folder
▼ GDL1BV1	Subfolder with machine name
HalconLibrary	Location of the image processing library
> Init	Configuration files of all BV scripts
▼ Machine	Machine-specific configuration files
ApplicationSettings	Configuration files of the software
MFL	Halcon script for multifocal lenses
> SPH	Halcon script for spherical lenses
> TOR	Halcon script for toric lenses
libs	Other programme libraries
▼ LocalSettings	Local Settings
Calibration	Location of the registration script
ImageComparing	Location for the script of the image comparison tool
Logging	Log system configuration
Python	Python script for ML
SmartCuvetteMemory	Clipboard for SCM
▼ Logs	Location of log files
Tuesday	-
Wednesday	-
Outbox	Outbox folder

Table 1: Folder structure Oklin3

After execution of the installation, a folder structure is created as shown in Table 1. All quality-relevant files can be found in the Inbox folder. The HalconLibrary subfolder contains the image processing library, which provides functions corresponding to the test scripts. The Machine folder contains the registration data for each camera head, a settings file with quality-relevant settings and files to load the camera configuration. In addition, there are backups of the last five registrations including change tracking with the date of the change and the user logged in at the time of the change. The ApplicationSettings subfolder contains all the files for the installation-specific configuration of the software that are created during the installation:

ApplicationSettings.conf	• General settings such as image storage format • SPS specific settings • Hardware configuration: Camera heads, cameras, LED controllers, number of lenses (number, names, etc.) • Configuration of the surface (WT and lens data to be displayed) • User groups • Configuration of statistical storage
CTHistoryMap.conf	List of available error classes for the WT history
ErrorMap.conf	List of available error classes for lens inspection
PLCMap.conf	Configuration of the interface to the PLC
ImageSetup.conf	Configuration for the image comparison tool

Table 2: Configuration files of the software

There is another subfolder per lens type that contains the associated Halcon script to perform the inspection. The script must contain the folder name (e.g.SPH) in the file name. If no or more files in the folder contain this part, no script is read in and there is the error "No test parameters" if this lens type is to be checked.

The LocalSettings folder contains files that are not quality relevant and need to be adapted from installation to installation, such as the configuration of the system's log output or settings for the registration of camera heads.

All system messages are stored in the subfolder Logs in text files sorted by weekdays.

All quality-relevant data, which are changed or generated by the user in the setup operation of the software, are stored in the Outbox folder. In order for this data to be loaded into automatic mode with the next start of the software or the next change, these must be transferred to the Inbox folder by the authorized users. The only exception to this concept is the functionality for the direct transfer of registration data into the inbox, which is available from the user level service.

3 Operation

3.1 Access Permissions

Different access permissions are provided for the operation of the software. These permissions are divided into three groups. The groups are to be classified in:

- NoUser (default user group without corresponding group in Active Directory™)
- Operator (operator)
- Service (maintenance)
- Administrator

Access to individual elements of the user interface is linked in certain places to the user group of the respective user. The user group NoUser has the least access rights and represents the default setting.

3.1.1 Notification

After starting the program, a user is automatically logged in as a user of the software. A user from the "NoUser" level does not have to log in separately to use the software.

Access to the software with extended access rights is only possible after a user has registered with the 5-2-1 identifier and his personal password. The login automatically determines the user group of the user and assigns corresponding access rights. If the user is not included in any group, he or she will be registered as 'NoUser'.

3.2 Interface

After starting the software, the default view of the program is displayed. This window is divided into different areas.

- The button bar in which different buttons for submenu can be found.
- The status bar where information of a general nature can be displayed.
- The main window showing the graphical elements that can be viewed by the operator during production.

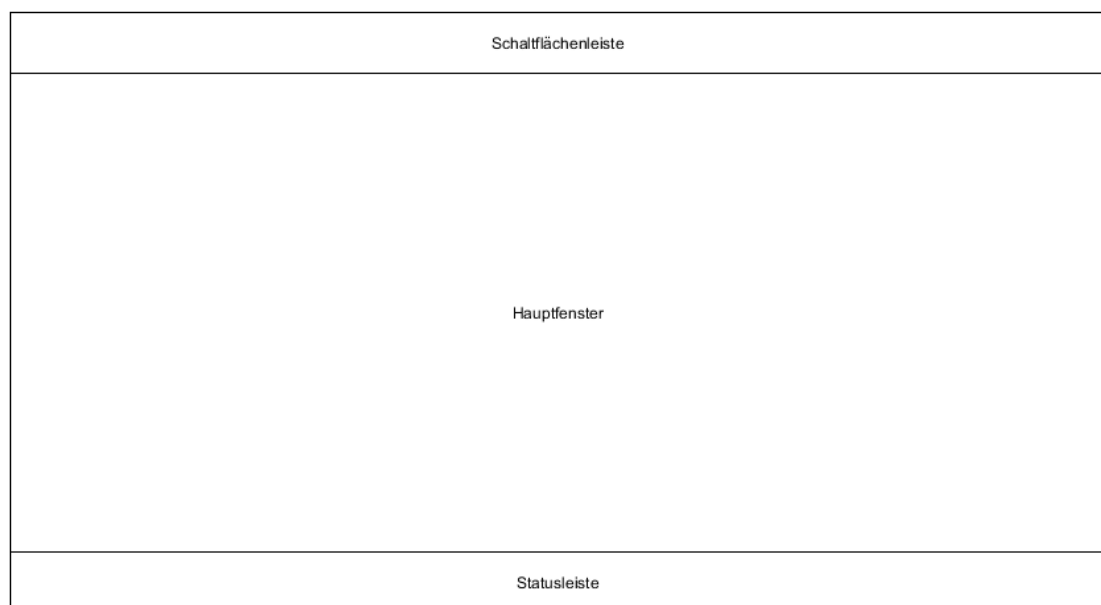


Figure 1: User Interface Areas

3.2.1 Main window

Depending on the system configuration, the structure of the main window differs with regard to the arrangement and availability of certain display elements. The following is an example of the main window.

The main window is divided into the following areas:

- 1 WT History
- 2 Operating elements / current WT data
- 3 Lens display
- 4 Variable display of additional information during operation

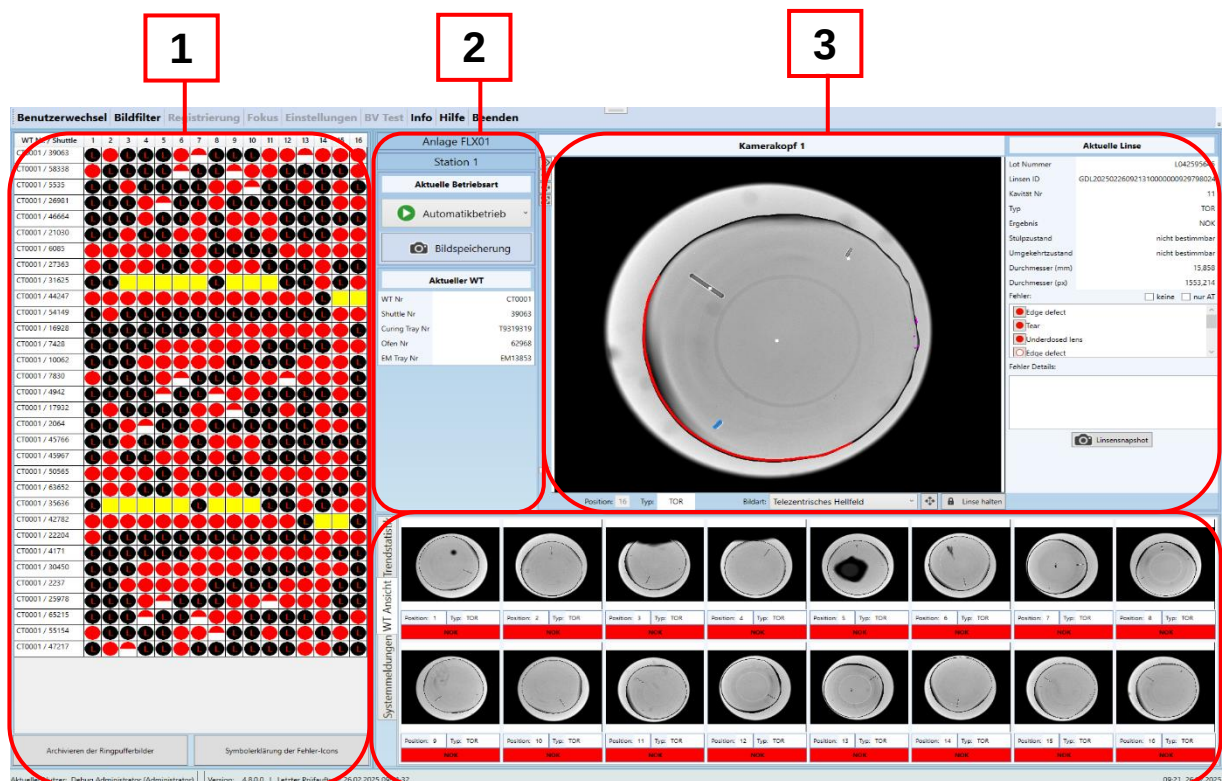


Figure 2: Main window with a camera head

3.2.1.1 WT History

In the WT history, lens results of the last WTs are visualized in tabular form using a certain error symbol. The number of WTs displayed in the WT history is automatically adjusted to the respective screen resolution. The first column corresponds to the entry "CTHistoryInfos" in ApplicationSettings.conf (e.g. ==References== WT No. or combination of WT No. and Shuttle No.), while the following correspond to the lens position within the respective WT.

Depending on the errors encountered, the lens positions are assigned to different main errors and symbols, which are specified during installation and stored in the Inbox folder in the CTHistoryMap.conf file. For example, for the LSFlex BV system type, the following defect symbols arise:

	OK
	OK (SCM corrected)
	KL not found
	Not inflated
	Illumination
	Recording error
	No test parameters
	No verification task
	Test error
	Error on lens

Table 3: Defect symbols

When clicking on a lens in the WT history, the images of this lens are loaded from the ring buffer and displayed on the GUI (Figure 4).

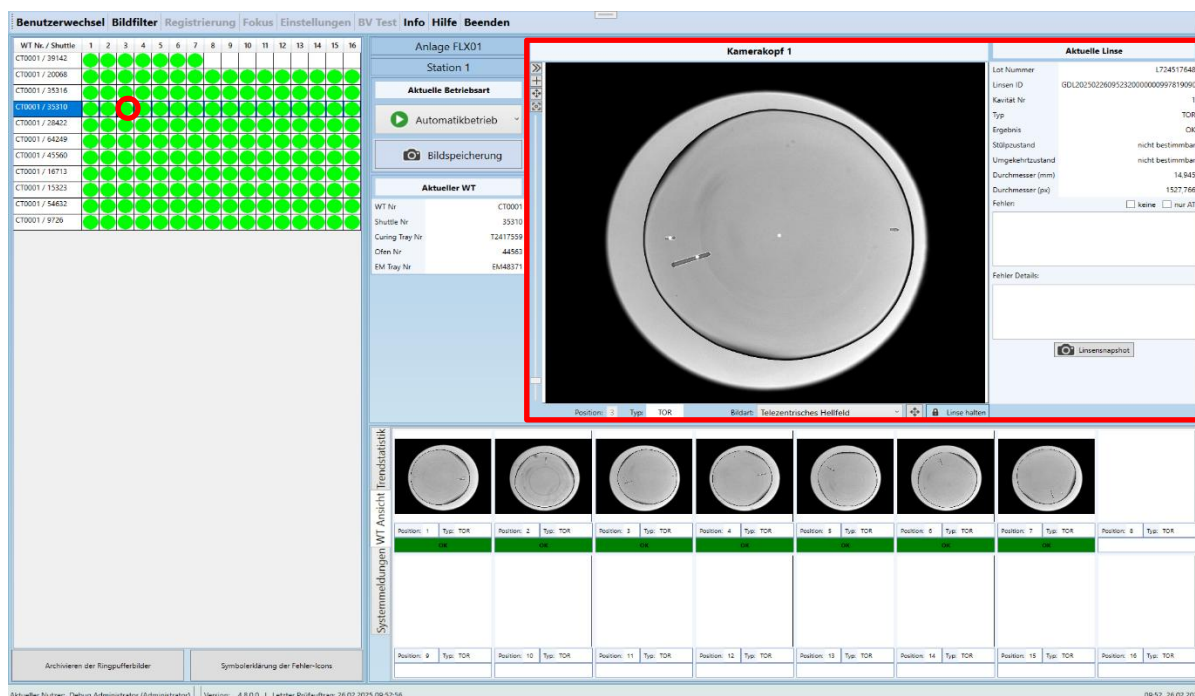


Figure 3: Ring buffer functionality. By mouse click on a position in the WT history (rounded in green on the left) the lens data belonging to this position and the corresponding image are loaded from the ring memory (red box on the right)

In addition to the images, the lens data, error list and error regions are loaded to the corresponding lens. In addition, it should be noted that the images of the ring buffer on the hard disk are stored in the original format according to the time of image capture, whereas the cached registration is applied to it when reloading. The number of WTs to be stored in the ring buffer can be set via the parameter `NumberCTHistoryInRingBuffer` in `ApplicationSettings.conf`. The default value 20 is taken if the parameter `NumberCTHistoryInRingBuffer` is not present in `ApplicationSettings.conf`.

When pressing the button Archiving the ring buffer images, the current images of the WT history from the ring buffer are permanently stored in the folder `D:/Oklin3/Ringbuffer_Archives`. This folder is not cleaned up when the software is finished.

By selecting the icon explanation button of the error icons, a window with the icon overview is opened.

Note: If the software is configured for offline operation as an evaluation tool, there are additional tabs in the area of WT history for the manual or automatic evaluation of stored lens images. Details can be found in a separate manual.

3.2.1.2 Camera head lens display

A lens display is divided into the following areas:

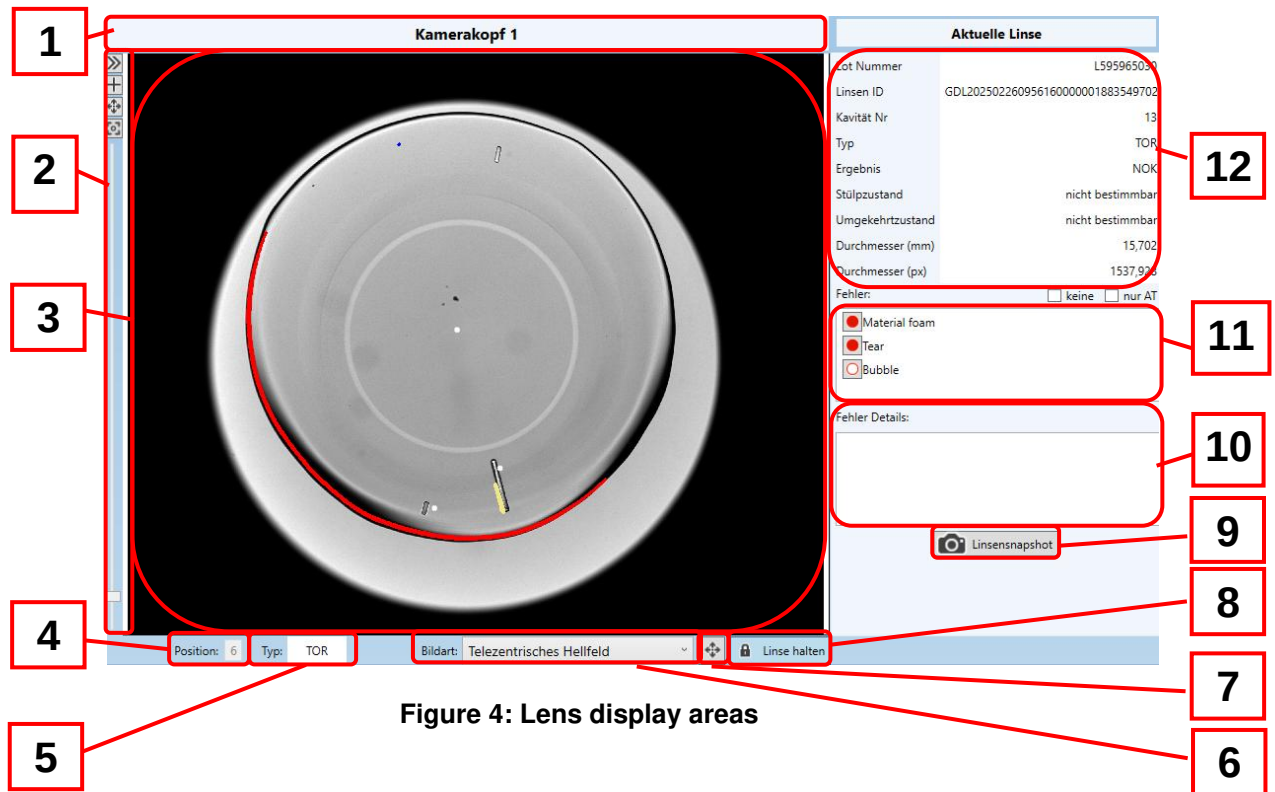


Figure 4: Lens display areas

- 1 Camera head, which has taken the lens
- 2 From top to bottom: Button to enable and disable the display on which the current mouse pointer position is displayed in image coordinates and the gray value at this position; a button to hide and hide a crosshairs in the image center as well as one; button to center the image and reset the zoom; button to focus a selected defect; zoom bar.
- 3 Recording image Images taken from the camera head are displayed here. Similarly, the defect regions found during the evaluation are drawn to the image in different colors depending on their defect class (see e.g. ==References== Table 4: Assignment of defect colors. The available defects as well as the associated colors are set during installation and stored in the file ErrorMap.conf in the Inbox folder). By left-clicking on such an error region, this can be selected and the corresponding results are automatically selected in the result display.
- 4 Position of the lens on the WT
- 5 Type of lens (plant specific, e.g. ==References== SPH, TOR or MUL for LS Flex BV)
- 6 Select the image type to display Here you can select the image type to display. Depending on the system configuration, at least one image type is available here. For LS Flex BV, for example, these are diffuse light field, dark field, phase contrast and telecentric light field.
The hotkeys F3 and F4 can be used to change the image type to be displayed: F3 iterates the dropdown forward, F4 iterates backwards. The function is rolling, i.e. after the last image the first one is selected again and the last one is selected after the first one.
- 7 Center lens When clicking the center lens button, the lens image is centered in the display.

- 8 Hold lens When activating the lens hold button, the currently displayed lens is held. New lenses will then no longer replace the image of the current lens until this function is deactivated again by re-activating it.
- 9 Lens snapshot: the images of the current lenses on the GUI can be saved (more about this: 3.2.4).
- 10 Error details: Tolerance (IT/AT), error class, defect size and position, duration of evaluation. In addition to ML scripts: Tolerance and error class of the top 3 results.
- 11 Results of the image analysis: A list of found defects is shown here. If a defect is selected, the corresponding region will be highlighted on the image and further details will be displayed in the text box next to it. Depending on the selection in the checkbox, "AT-only" displays all errors or only AT-class errors. With the checkbox "no" the errors can be completely hidden. The checkbox can also be selected or selected via F5.

Blue	Bubble
Orange	Star Crack
Magenta	Edge error
Pink	Bump with crosslinking
Red	Crack
Gold	Roadmaps
Light turquoise	Inclusion
Azure blue	Under-dosed KL
Coral red	Surface defects
Kaki	Material foam
Turquoise	Printing errors
Pink	Pseudo errors
LightSeaGreen	ML pseudo error (currently only with LS Flex2+)

Table 4: Assignment of defect colors on the example of LS Flex BV

- 12 Lens data. Depending on the system configuration, the position may vary.

3.2.1.2.1 Image display/photograph

Lenses are displayed by means of an image display (see Figure 5). In this image display, the image can be moved and scaled (mouse wheel) with the help of the mouse. In addition, scaling is also possible with the zoom bar, which is arranged to the left of the image. There are three buttons above this bar. The upper button activates and disables an display that displays the current mouse pointer position in image coordinates and the gray value at this position.

The middle button activates or disables a red crosshair to visualize the center of the image. If a lens with a script was evaluated for the given display window, which writes the center of the bottom lens in the form of offsets relative to the image center in the feature list (Features "BottomLensOffsetX- and "BottomLensOffsetY-), a blue crosshair is also drawn, rotated by 45° (Figure 6).

The lower button centered and scaled the image to the original state.

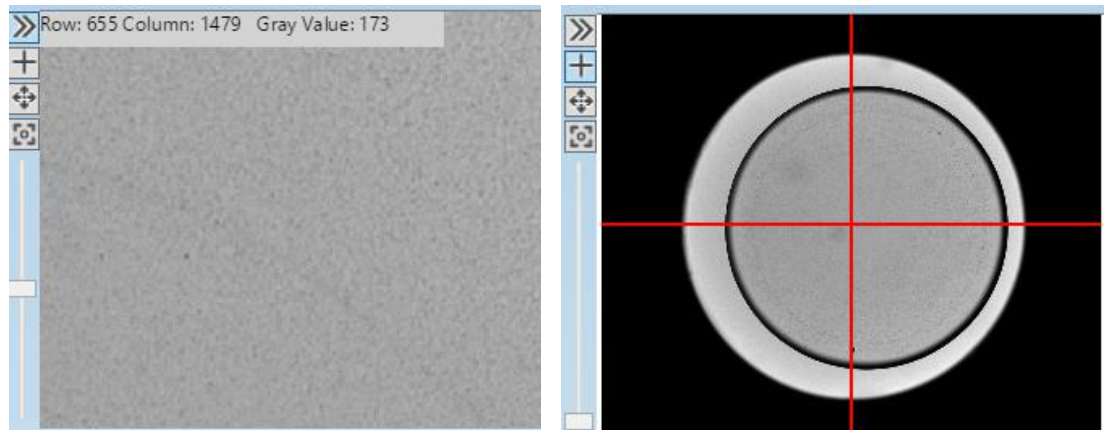


Figure 5: Image display with zoom bar and gray value display (left) and Crosshairs (right)

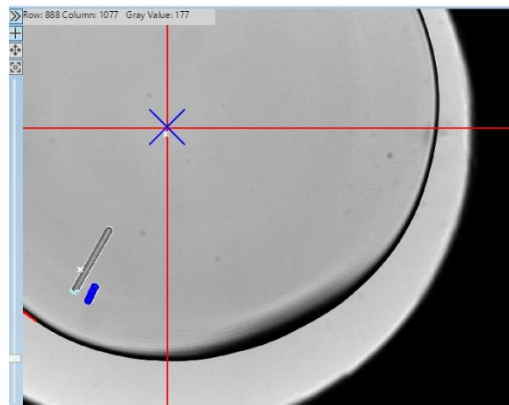


Figure 6: Bodenlinsenzentrum as a blue crosshair

3.2.1.3 Operating elements / WT information

Standby display

If the system is ready for production, this is indicated by a green footprint.

Change Mode

If the corresponding permission of the currently logged-in user allows this, the following menu can be used to manually switch between automatic operation and set-up operation.



Figure 7: Mode of operation Change

Further information on the operating modes can be found in the section Operating modes 3.3.

The current WT data, which were transmitted by the PLC, are always displayed in the "Current WT" range. Which of the available data is displayed on the surface is specified during installation and stored in the file ApplicationSettings.conf.

Aktueller WT	
WT Nr:	dkwoshgoizvaaaaa
Mold Nr:	677
UV Bay ID:	458
Extraktions Nr:	EX1234567890a599

Figure 8: WT Information Display

If the image recordings of the camera heads should be stored permanently during operation, the image storage can be activated via the following button:

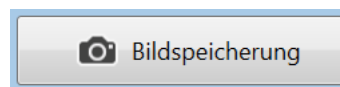


Figure 9: Image Storage Button

Options for how to configure image storage are discussed in Picture Filters, page 19. During active image storage, the button is bordered red:



Figure 10: Image storage button with active image storage

After exiting active image storage or missing image storage information in the image filter window, the button is locked:

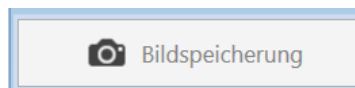


Figure 11: Image storage button locked

3.2.1.4 Installation information

The information of the installation is displayed on the GUI. By default, the installation name and the station index are displayed here (Figure 12).

Anlage GDL10
Station 1

Figure 12: Appendix name and station index on GUI

For the high-cavity systems, the High-cavity module index is also displayed:

Anlage HCL01
Modul 2
Station 1

Figure 13: Display of system information on a high-cavity system

3.2.1.5 Additional information during operation

It is possible to switch between different additional information displays.
This is done by selecting the desired tab on the left side in this area.

By default, the Yield display is activated here.

Yield display

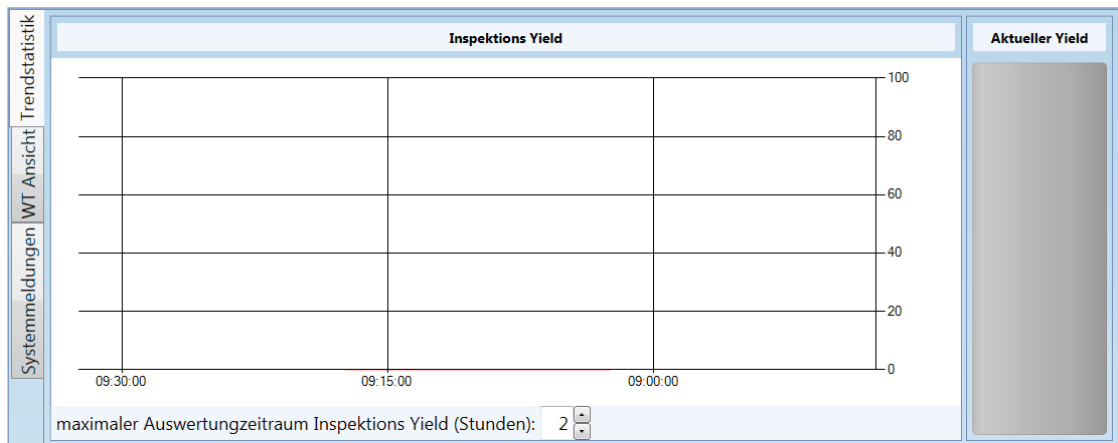


Figure 14: Trend statistics for cosmetic inspection

The Yield display displays the Yield history by means of a line diagram. The maximum evaluation period in hours is adjustable below the diagram. Depending on the system configuration, the current Yield value is also displayed on a red percent bar to the right.

For the calculation of the values, the moving average of the last 10 WT's is determined. The ads are updated for each WT after receiving the results.

The following formula is used to calculate the yield of the cosmetic inspection:

Inspektions – Yield

$$= \frac{\text{Number of KL with result class "Good"}}{\text{number of KL with test job and existing test parameters}} - \frac{\text{number of KL with result class "No-KL"}}{\text{number of KL with test job and existing test parameters}}$$

In the case of the configuration of an Inverted Lens Detection (ILD), the trend statistics, on the other hand, show the percentage of inserted and inverted contact lenses:

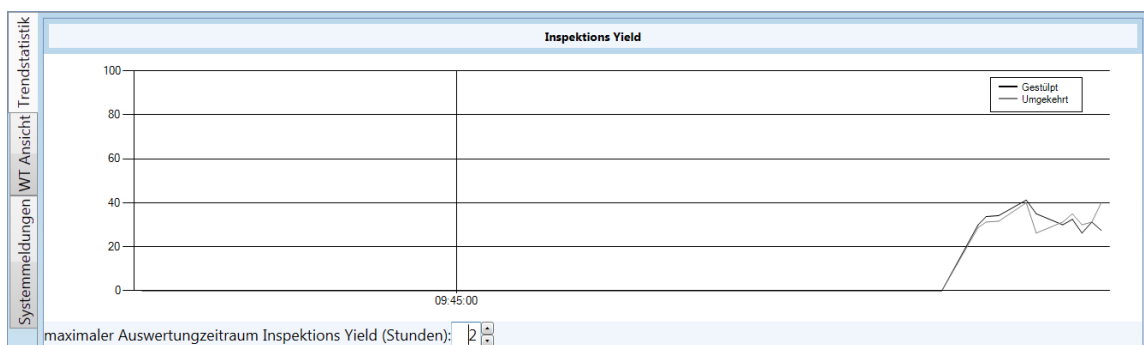


Figure 15: Trend statistics of an ILD

WT View

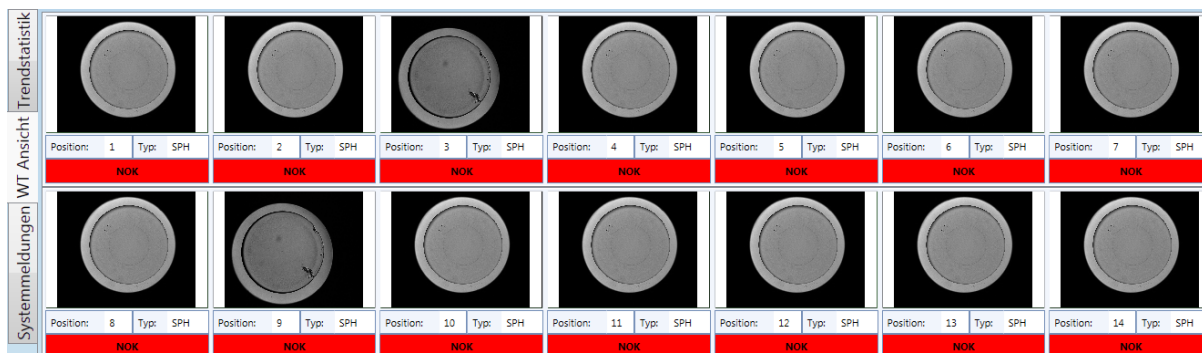


Figure 16: WT view for cosmetic inspection

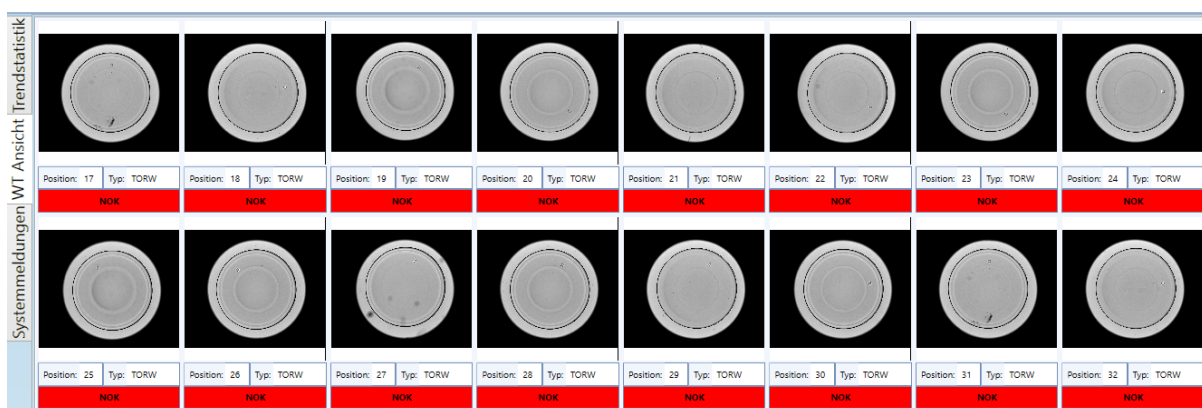


Figure 17: WT view for cosmetic inspection on the second high-cavity module

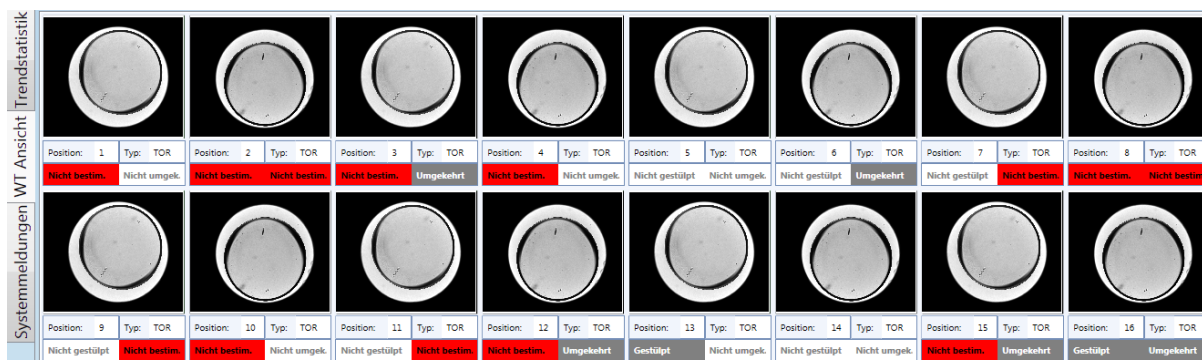


Figure 18: WT View for ILD

In the WT view all lenses of the current WT are displayed. In addition to lens position and lens type, the final lens evaluation is displayed for each lens. In cosmetic inspection, these are "OK" and "NOK" respectively. In the case of an ILD, the left half of the stud state ("Stülpt", "Not stülpt" or "Not determinate") and the right half of the reverse state ("Reverse", "Not vice versa" or "Not determinate") are displayed.

Here you can also directly select a lens with the left mouse clicks to hold in the lens display.

System messages

Internal messages of the program flow are displayed within the system messages.

Each message is preceded by timestamp and log level. The system messages are additionally stored in log files. This storage takes place according to weekdays sorted in the folder "Logs", which is parallel to the executable file.

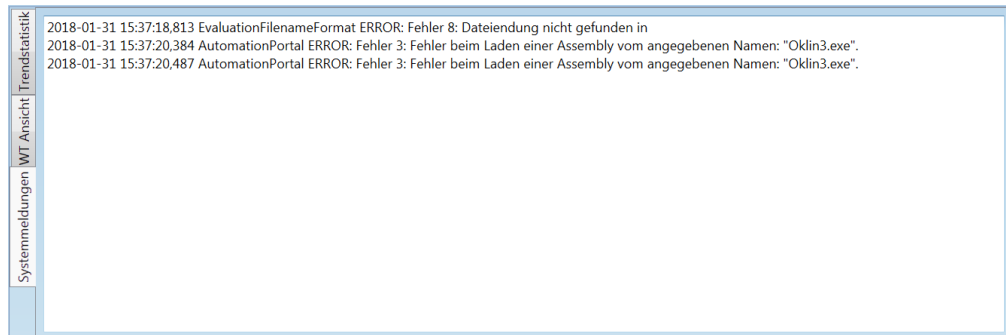


Figure 19: System messages

3.2.2 User login

If a change of the user is initiated by pressing the button "user change", the user name and password must be entered via the following dialog.

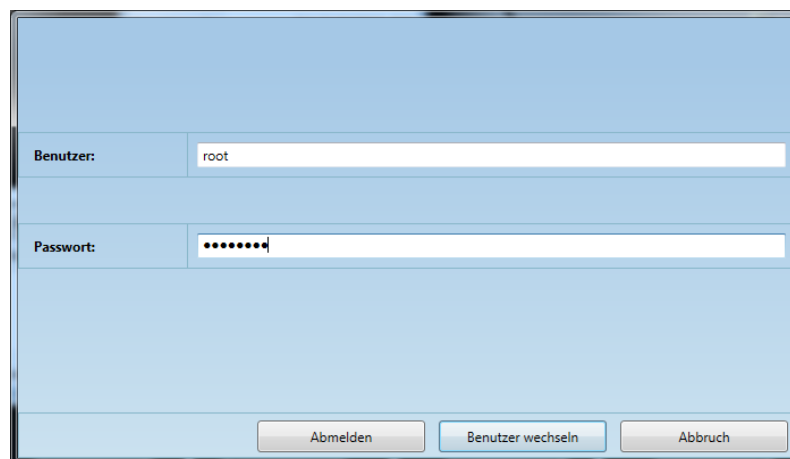


Figure 20: Login dialog

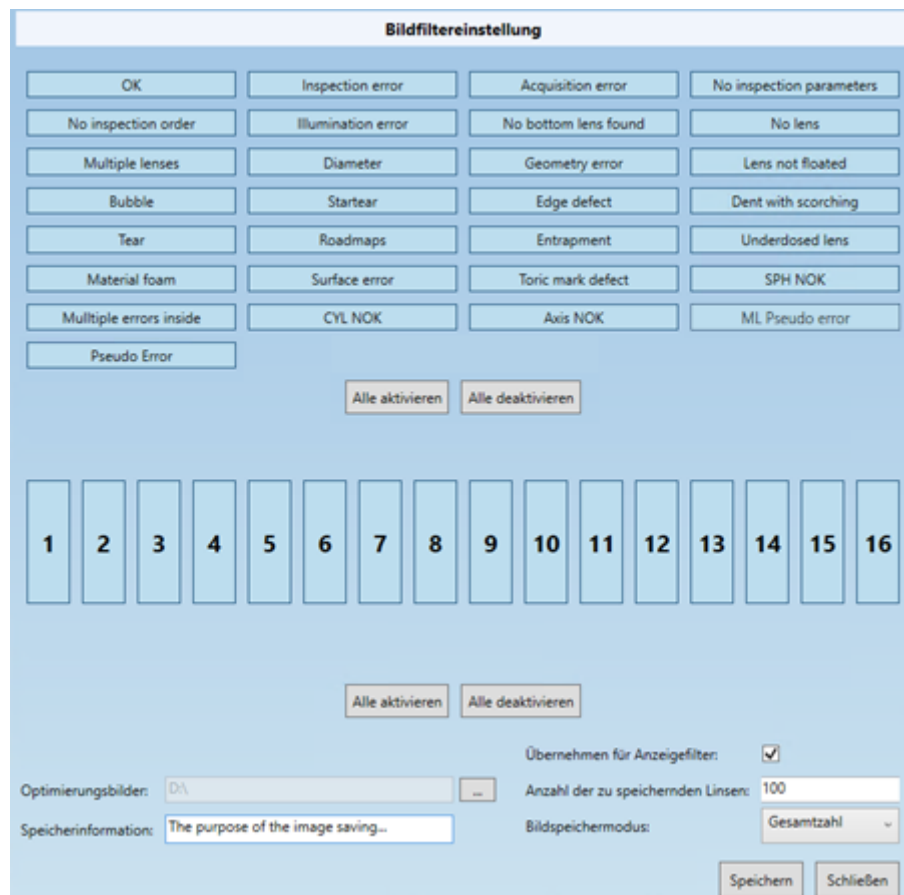
The user's authentication is done via the Microsoft Active Directory™ service. Based on the group affiliation of the authenticated user, the permissions within Oklin3 are granted.

3.2.3 Image Filter

The image filter setting refers to the image storage for optimization purposes, which can be activated in the main screen. If a tick is set for the Checkbox Apply for display filters, the image filter setting is also used for the display of the images in the large windows on the main view.

It is possible to activate both types of errors and positions within the WT for filtering. Furthermore, the location and number of images to be stored are adjustable. Storage on hard disk C is not permitted. Depending on the system configuration, different fields are created for selecting errors and positions.

The following diagram shows the window example for LS Flex 2+ BV:



Bildfiltereinstellung

OK	Inspection error	Acquisition error	No inspection parameters
No inspection order	Illumination error	No bottom lens found	No lens
Multiple lenses	Diameter	Geometry error	Lens not floated
Bubble	Startear	Edge defect	Dent with scorching
Tear	Roadmaps	Entrapment	Underdosed lens
Material foam	Surface error	Toric mark defect	SPH NOK
Multiple errors inside	CYL NOK	Axis NOK	ML Pseudo error
Pseudo Error			

Alle aktivieren Alle deaktivieren

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

Alle aktivieren Alle deaktivieren

Optimierungsbilder: D:\ ...

Speicherinformation: The purpose of the image saving...

Übernehmen für Anzeigefilter: ☒

Anzahl der zu speichernden Linsen: 100

Bildspeichermodus: Gesamtzahl

Speichern Schließen

Figure 21: Image filter setting dialog on the example of LS Flex 2+ BV

All settings relevant to image storage are loaded from the localSettings.conf file in the LocalSettings folder when opening the window or discarding unsaved changes. When saving, the file is updated accordingly.

Image storage mode:

It can be selected between three different counting methods for image storage:

- Total number: The adjustable number of lenses to be stored refers to all stored lenses
- Per position: The image storage runs until the set number of lenses is stored per position selected (via the filter).
- Per error: The image storage runs solagne until the set number of lenses is reached each (defect class selected via the filter).

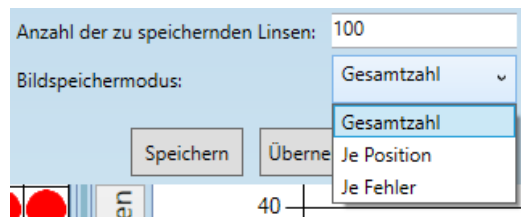


Figure 22: Select Image Storage Mode

Storage information:

Before an image storage by pressing the image memory button or in the case of a already active image storage by saving or taking over the new Image filter settings can be started, a description for the reason of the Storage in the "Storage information" field. The field is on 50 characters and when using Tif images in the field "Subject to the Image metadata written. The field is automatically deleted when image storage end (either because the user is saved by the image memory button aborts or because one of the end criteria for storage is met).

Recurring image storage:

For the default user of the level "NoUser" only the view from Figure 21 is visible, for users of higher user levels (from user level "User") there is also an extended image storage available:

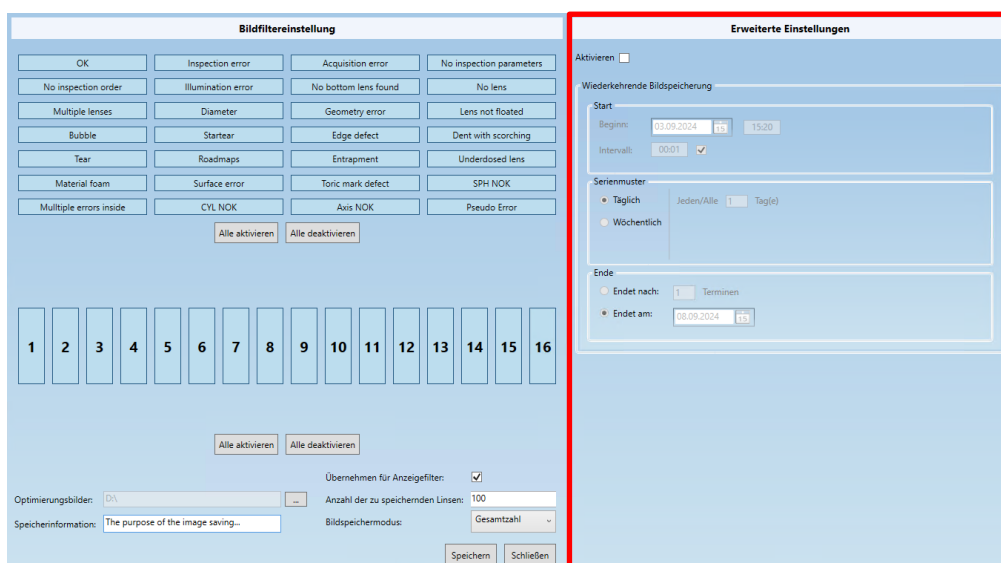


Figure 23: Advanced Image Storage

In order for the recurring image storage to be edited and activated, it must be explicitly selected via the check mark "Activate". If an operator wants to change the image storage under the default user "NoUser" while active image storage is already configured, it will be informed via a pop-up window. If it confirms the acquisition, the recurring image storage in the background is automatically deactivated.

The configuration of recurring image storage is divided into three sections:

- **Start:** Here the start date and start time of the recurring image storage series can be set. Optionally, an interval storage and the double distance between two intervals can be adjusted.
- **Model series:** Here you can configure on which days the image storage should be active. It can be selected between daily and weekly settings.
 - **Daily:** It can be set that the image storage should be activated every X days from the start date and time for 24h each (Figure 24).
 - **Weekly:** It can be set that the image storage is activated every X weeks from the week in which the start date is held, on the selected days of the week, for 24 hours each (Figure 25).
- **End:** Here the demolition criterion of the series can be defined
 - **Ends:** The end date determines the end of the series. The recurring image storage will be terminated at the selected date when the selected start time is exceeded.
 - **Ends after:** Here the number of image storage events (not intervals) can be entered, after which the series is to be canceled.

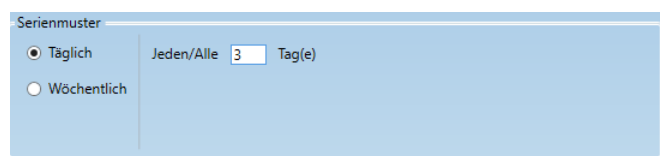


Figure 24: Daily serial pattern. From the selected starting point, the image storage in this example active every three days for 24h

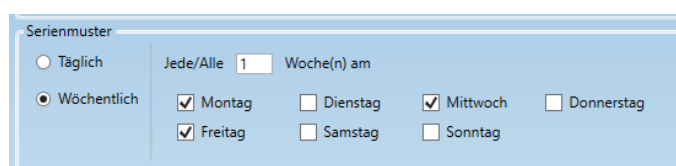


Figure 25: Weekly serial pattern. In the example shown, the storage of each week on Mondays, Wednesdays and Fridays active for 24h each

The optional interval storage makes it possible to activate the storage in equidistant times within an active storage event. For example, an interval of 08:00 means that a new storage is initiated every eight hours on a day when this storage is active.

For recurring image storage without interval storage, an image folder consisting of the time stamp of the start time of the event and the event counter is created for each new event, e.g. B.:

 2023-05-03_10-53-00_e0001

Figure 26: Exemplary subfolder for recurring image storage without Interval storage

The time stamp of the interval start point as well as the interval counter is used for activated interval storage, e.g. B.:

 2023-05-03_14-48-00_i0007
 2023-05-03_14-51-00_i0008
 2023-05-03_14-54-00_i0009

Figure 27: Exemplary subfolders for recurring image storage with interval storage. An interval of three minutes was set

Important: The times and event or interval counters always refer to the configured start time. This means that when a series with starting time is continued in the past, the times and counters are continued as if the series had never been interrupted. However, this does not apply to the picture counters. These are continuous within a running, recurring image storage, but after stopping a series the counters are reset.

The recurring image storage is linked to the filters regarding error class or cuvette position via AND.

Abortion of image storage:

The storage is interrupted in the following situations:

- The user breaks the storage manually by pressing the image memory button. The selected number of lenses has been saved
- Recurring image storage:
 - The final criterion of the series was reached (either the last event of the series was passed or the final date (taking into account the start time) was exceeded). It is
 - the last event active and the required number of lenses was saved before reaching the end time

Move the pictures from the ring buffer:

For performance reasons, the images are not stored directly in the target directory, but are initially kept in the folder for the ring buffer of the WT history. The images marked for the optimization image storage are moved from the ring buffer to the selected image storage folder in the following situations:

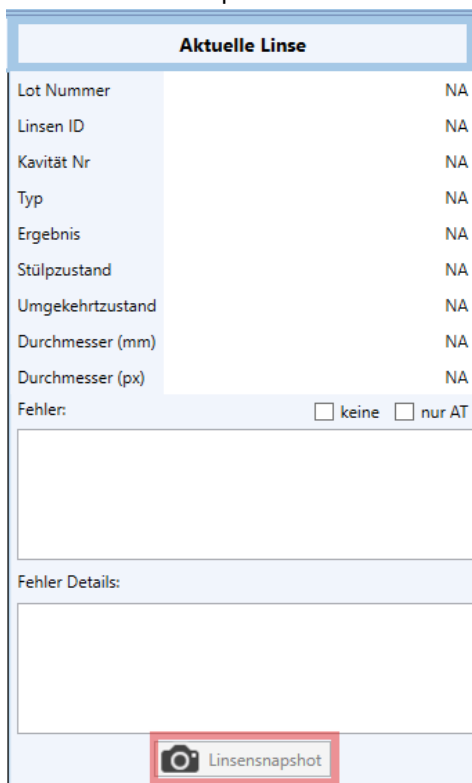
- Finishing image storage by pressing the image storage button
- Unique image storage: The selected number of lenses to store has been achieved
- Recurring image storage
 - The end of the series has been reached
 - The end of an active event has been reached
 - If interval storage is active: The end of an interval has been reached

- The last event/interval is active and the set number of lenses has already been reached.

When pressing the button for archiving the ring buffer, all images of the ring buffer are immediately moved to the intended folder. In this case, the images cannot be moved to the folder for optimization image storage.

3.2.4 Lens snapshot

There is also the possibility to save the lens currently displayed on the Oklin3 GUI via the button "Linsensnapshot" .



Aktuelle Linse	
Lot Nummer	NA
Linsen ID	NA
Kavität Nr	NA
Typ	NA
Ergebnis	NA
Stülpzustand	NA
Umgekehrzustand	NA
Durchmesser (mm)	NA
Durchmesser (px)	NA

Fehler: ☐ keine ☐ nur AT

Fehler Details:

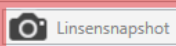
 Linsensnapshot

Figure 28: Lens snapshot button

The default memory path is: 的D:\Oklin3\Snapshot, where an additional subfolder in the format YYYY-MM-DD is created during the storage process.

The name of the saved image corresponds to the name configured in the ApplicationSettings.conf. The image is saved with metadata in the same way as with automatic image storage. This snapshot function works both for the last lens displayed and for a held lens (either via the WT history (ring buffer), the "hold lens" button or the WT view selection).

In automatic mode, the snapshot button remains disabled for a held lens after saving the snapshot until another lens is selected or the "hold lens" button is released.

For the setup mode:

- The last image displayed is always saved, even if it is a "Dummy" image with an error message, e.g. because the image was no longer available when loading from the ring buffer.
- The snapshot button is deactivated as soon as one of the setup operating windows (registration, focus, BV test) is opened. The button is activated again as soon as a lens is loaded from the ring buffer via the WT history.

The lens snapshot function is not possible in the evaluation tool.

3.3 Types of operation

The test system can be operated in two different operating modes. Both modes of operation are described in more detail below.

3.3.1 Automatic operation

This mode of operation is the standard mode of operation. It can only be produced in automatic mode. The control of the software is largely taken over by the control of the IM. The operator only has the option to set image storage parameters (image storage for optimization purposes), as well as to start or stop image storage.

3.3.2 Setting-up operation

If there is no production, the system can be operated in set-up mode. Switching to this mode is only possible with certain access rights.

The setup mode is used to configure the system. Users equipped with appropriate access rights can set and secure all changeable parameters of the system.

3.4 Version information

By pressing the button "Info", different version information can be viewed

Parameter Version

Skripte

Produkt	Version	Permanente Winkelmessung	SCM Aktiv	ML Aktiv
MFL FT32	2.6.0	inaktiv	N/A	N/A
SPH FT32	2.6.0	inaktiv	N/A	N/A
TOR HEVL	3.0.0	inaktiv	N/A	aktiv

Bibliothek

Bibliotheksversion: 107

Machine Learning

ML ist aktiv

Modell-ID: xt6e

SmartCuvetteMemory

SCM ist nicht aktiv

Moldmarkprüfung

Look-Up-Tabelle	Version
ToricLookUpTable_Axis.csv	1.0
ToricLookUpTable_Cylinder.csv	1.0
ToricLookUpTable_Diopter.csv	1.0

Datum der letzten Registrierung

2020-02-06 09-02

Software Version

5.0.0.0

OK

Figure 29: Version information

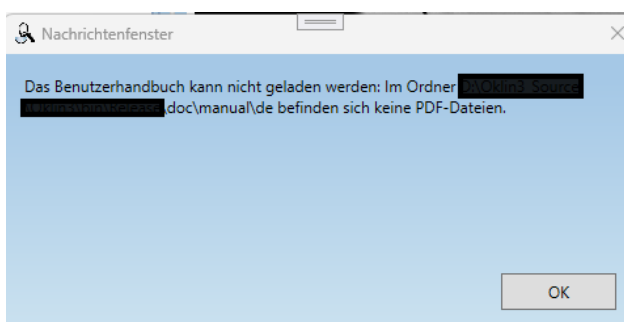
3.5 Help

The current Oklin3 manual can be opened via the "Help" button. To open the manual, the same PDF viewer program as the PDF report is used for the hardware jig setup or the default viewer set in the system.



In the Oklin3 installation directory (usually C:\MA\Oklin3) there is a folder "doc" with another subfolder "manual" which contains a subdirectory for each supported language ("de" for German and "en" for English). In each subdirectory there is a PDF file with the Oklin3 manual in the respective language.

If the folder or PDF file is missing for the configured language, an error message is displayed when clicking the Help button.



If there is more than one PDF file in the corresponding folder, a warning is logged in the Oklin3 log file when the Help button is clicked. In this case, the first PDF file in the directory is selected.

4 Advanced functions

The extended functions serve the purpose of configuration and maintenance of the system, which are reserved for the user groups operator, service and administrator and can only be accessed in set-up mode.

Changes to machine parameters are stored in the outbox. When switching to automatic mode, the original machine parameters are read from the inbox. In order to take over the machine parameters for automatic operation effectively, these must be taken over from the outbox via the SSRS into the inbox. The SSRS creates an audit trail and versions the machine parameters. The implementation of the SSRS is not part of Oklin3 and is the responsibility of the operator.

By right-clicking on the camera name, it is possible to display the current images in a separate window in each window in the setup mode.

4.1 Registration

The purpose of the registration depends on the type of installation. When using four cameras per camera head, it is necessary to determine the deviation of the camera images from the master camera. The calculated result is a corrective transformation for the respective cameras, which takes translation, rotation and scaling into account. These transformations are then automatically provided for lens analysis images. Furthermore, the registration serves to determine the image scale in μm per pixel. For the configuration of one camera per camera head, however, the transformations are not determined.

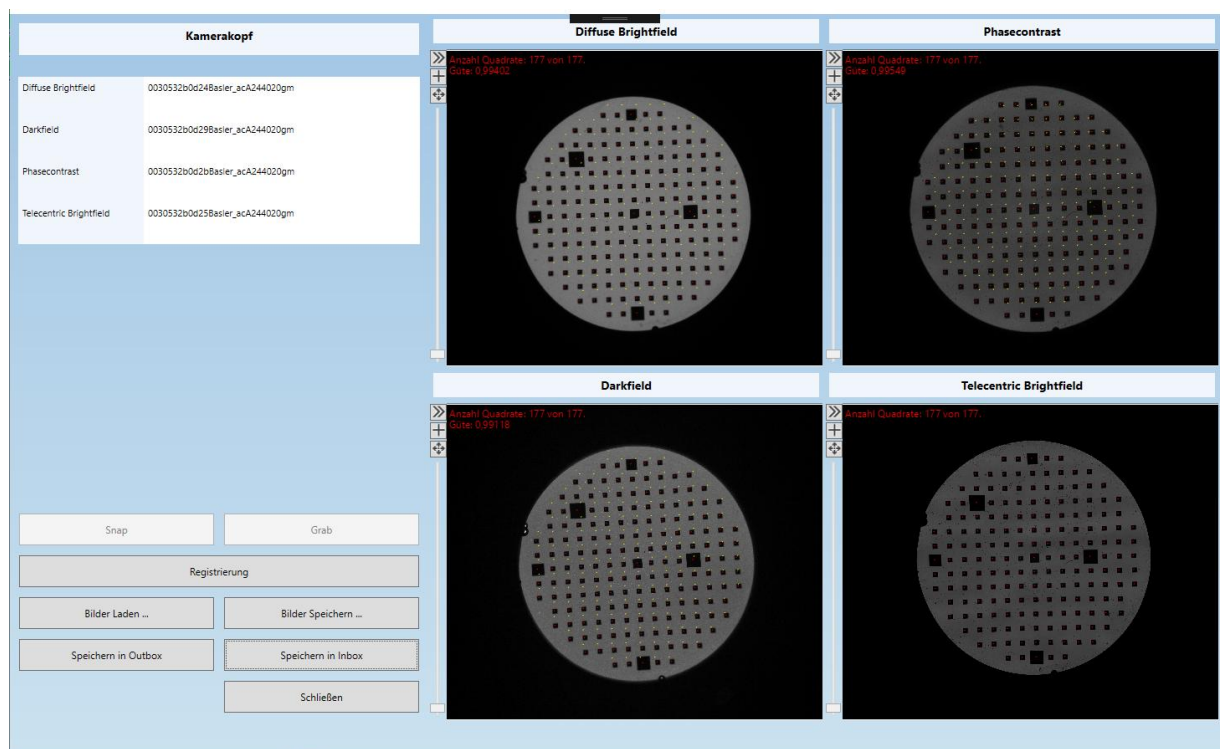


Figure 30: Camera registration dialog with 4 cameras per head

The following steps shall be performed for each of the camera heads:

- (1) Insert the registrar cuvette filled with water
- (2) Create recordings via the Snap button
- (3) Press Registration button

In the case of dark-field lighting systems, a device must also be installed before the image is taken. Diffuser to be placed on the lighting.

Via the button Images Download you can invite previously saved registration images and. Both loaded images as well as images taken via snap or grave can be registered via the Registration button.

The Save Images button opens a browser dialog where a file name and a Store location to be selected. All images will then be displayed according to the Installation specified image format saved.

There are two options for transferring the registration data into the Inbox. On the one hand, the data can be stored in the Outbox folder via the Save in Outbox button. The data are stored in the database. The data can then be entered in the inbox as described above. be postponed. On the other hand, it is possible to add the registration file directly to the Inbox folder. Write where this button is activated only from the service level. After confirmation of the Security query (see Figure 31) in this case is the currently used Registration file moved to a Zip archive in the Machine folder along with a Text file indicating the time of the change as well as the currently registered user log, as illustrated in Figure 32. There will always be the last five Keep archives, the oldest one will be deleted if this number is exceeded.

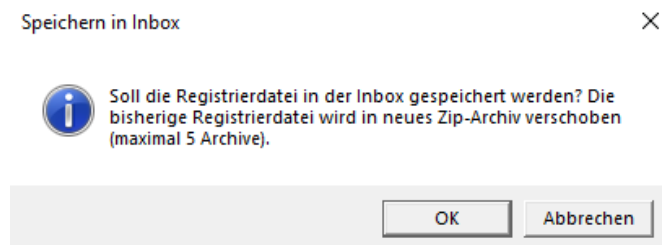


Figure 31: Security query for the direct transfer of registration data into the Inbox

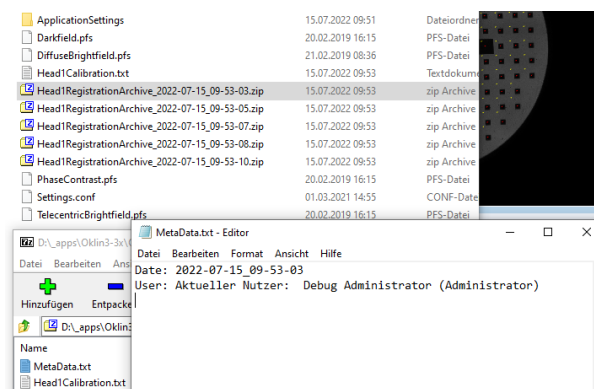


Figure 32: Archiving of registration files

4.2 Focus Check

Within the focus test, the camera systems of the IM systems can be adjusted as well as their images can be compared with each other. The "Camera System" tab is available for setting the cameras. There are also four more riders for different tests: "General", "Linse", "Focus+Resolution" and "Lighting." Depending on the system configuration, the window differs with respect to certain display elements. The following pictures show the focus window for the configuration of a camera head with four cameras.

Note: While the grave button is active in a tab, switching to another tab is not possible.

Note: There is a separate display for each camera. An exception here is the ILDi system. In this system there is a camera that can trigger two different lightings. There is a separate display for each lighting. Thus, there must also be two "Snap" and two "Grab" buttons in each tab, as only one lighting is used at the same time.

4.2.1 Camera system

The camera system is used to assign cameras to certain image types (e.g. telecentric light field, dark field, diffuse light field or phase contrast for DSM Flex) and to change camera parameters. Depending on the system configuration, the display elements of this window differ in terms of the number of camera heads and cameras per head and the image names of the LED controller channels per camera. The following figure shows the example window for DSM Flex BV:

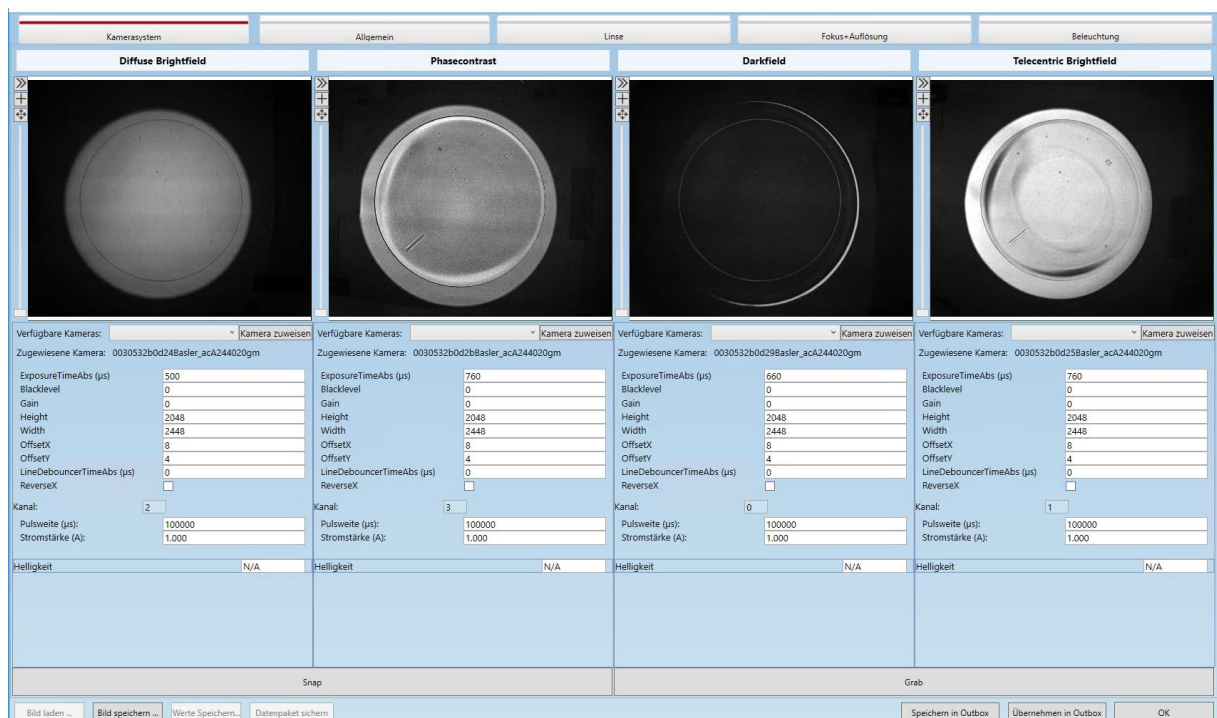


Figure 33: Camera system configuration using the example of DSMFlex BV

4.2.1.1 General

Choices in the camera system exemplary for LS Flex BV:

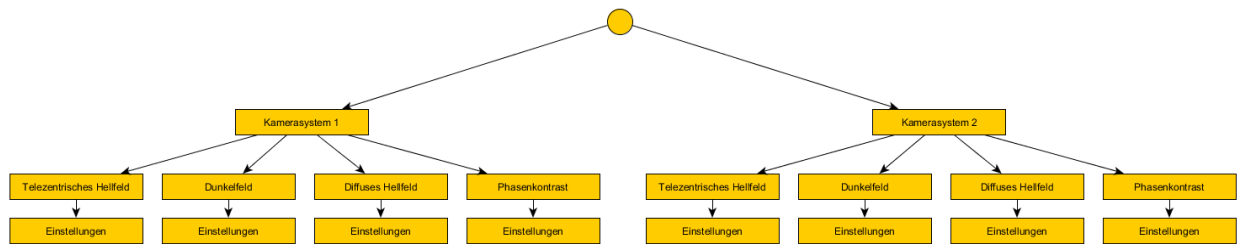


Figure 34: Choices Camera system using the example of LS Flex BV

The names of the cameras for display in the camera system window are specified during installation. Within the software, cameras are clearly identified via their MAC address.

4.2.1.2 Allocation of a new camera in the system

Allocation of camera parameters and camera

- (1) Choosing the camera system in the software (camera head if the number of heads is greater than 1)
- (2) Selection of one of the camera designations of the selected camera head (e.g. telecentric light field)
- (3) Choosing the appropriate MAC address from the list (dropdown menu to "Assigned Camera")
- (4) Button "Assign Camera"
- (5) Now the selected camera is connected to the software and camera parameters can be changed

4.2.1.3 Customize camera system values

Before shooting (snap or grave) in the camera system, the values shown on the surface are transmitted to the camera.

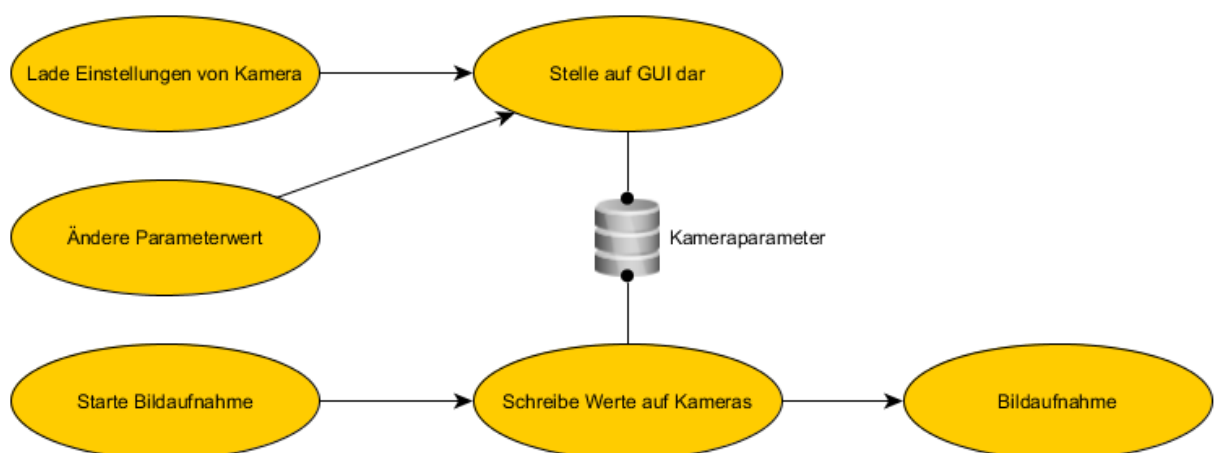


Figure 35: Parametrization Camera

4.2.1.4 Save Images

The Save Images button opens a browser dialog, in which a file name and a storage location are selected. All images are then saved according to the image format specified during the installation.

4.2.1.5 Backup Settings

When saving or taking over, the values displayed on the surface are transmitted to the camera and stored in the outbox.

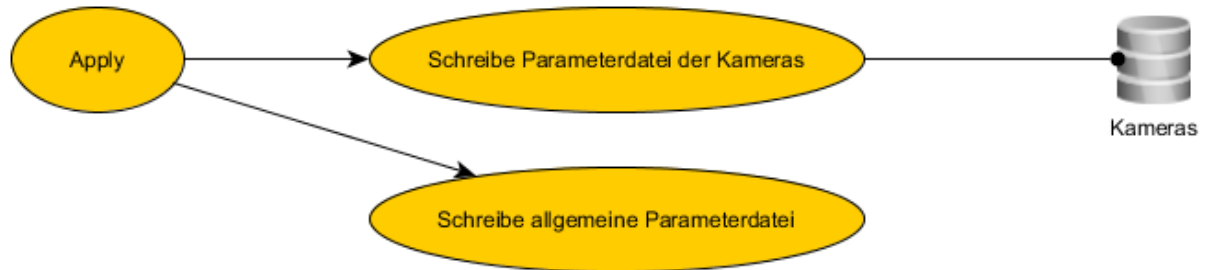


Figure 36: Back up camera settings

4.2.2 General

In this tab, the brightness along the red horizontal and vertical lines is calculated by the center point of the current image section. When moving the image (also scaling), these lines are recalculated as the image section has changed.

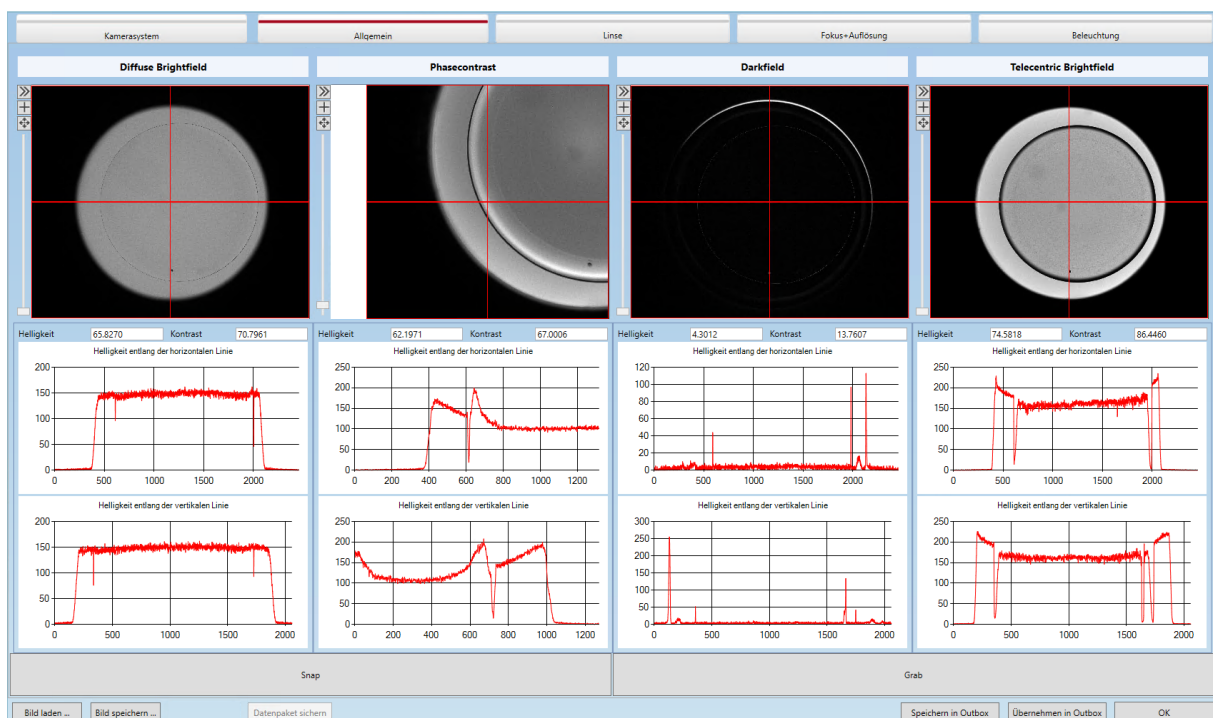


Figure 37: Brightness test using the example of DSM Flex BV

After selecting the camera head to be checked, a single image (snap) of the cameras can be started to start the evaluation.

A previously saved image can be invited via the Image Load button.

A browser dialog opens via the Save Image button, in which a file name and a storage location are selected. All images are then saved according to the image format specified during installation (see 4.5.5).

4.2.3 Lens

The "Linse" tab (Figure 39) is only available if the two files are present:

- *"ImageSetup.conf" in the directory "..\Inbox\[machine name]\ApplicationSettings" and*
- *"ImageComparingTool.hdev" in the directory "..\LocalSettings\ImageComparing"* at the start of the software.

With the option of "Snap" or "Images Load", the Halcon script is executed in the background, which calculates the corresponding values, which are then displayed and colored (green=good, yellow=still OK, red = not OK).

The respective limits of the values for determining the background color are read from the configuration file ImageSetup.conf. Figure 38 shows an example of the setting to determine the background color. The Brightness value is not displayed in the Telecentric Brightfield, Diffuse Brightfield and Phase Contrast Camera (IsDisable=False) and in the DarkField Camera (IsDisable=True) (Figure 39). ==See also== "Brightness*Brightness*Brightness*Brightness*Brightness*Brightness*Brightness*Brightness*Brightness*Brightness==References== For example, the Telecentric Brightfield camera is marked differently:

- If it lies between OptimumMin (152) and OptimumMax (162) (so-called optimal values), it is marked green. The borders are included.
- If it is between Min (130) and Max (190) (so-called acceptable values), it is marked yellow. Again, the borders are included.
- If it is smaller (130) or larger (190), it is marked red.

The optimal and acceptable values for each result value are displayed as a tooltip if the mouse pointer is above the text field.

```

,,
"CompareParameters": [
{
  "Name": "Brightness",
  "HalconOutputParameter": "OpticZoneBrightnessValues",
  "Values": [
    {
      "Image": "Telecentric Brightfield",
      "Min": 130.0,
      "OptimumMin": 152.0,
      "OptimumMax": 162.0,
      "Max": 190.0,
      "IsDisabled": false
    },
    {
      "Image": "Diffuse Brightfield",
      "Min": 140.0,
      "OptimumMin": 150.0,
      "OptimumMax": 162.0,
      "Max": 172.0,
      "IsDisabled": false
    },
    {
      "Image": "Phasecontrast",
      "Min": 81.0,
      "OptimumMin": 96.0,
      "OptimumMax": 106.0,
      "Max": 121.0,
      "IsDisabled": false
    },
    {
      "Image": "Darkfield",
      "Min": 0.0,
      "OptimumMin": 1.0,
      "OptimumMax": 2.0,
      "Max": 3.0,
      "IsDisabled": true
    }
  ]
}

```

Figure 38: Example for the configuration of the background color

If the ImageSetup.conf file contains incorrect entries or if the comparison parameters check fails, the tabs will be disabled. In this case, a system message will be sent. Typical example of a wrong entry is if the optimal limits are not between Min and Max.

The calculated values can be saved in a text file via the button "Save Values... For traceability, the name of this text file also includes the current date and time and the name of the inbox in the form of "Date_Time_Stamp_{Head Index}_{Inbox name}.txt".

All values not yet calculated are specified in the text file with n/a. The script usually returns "NA" for parameters that are not returned by the script.

In addition, there is a button "Grab" that records the current image in live mode and performs the calculation of the checks cyclically.

The images used can be backed up via the button "Save Images...".

In the "Linse" tab there is the possibility to perform the evaluation with changed input parameters and evaluation limits (Figure 39). These must also be specified in the configuration file. The following tabs are provided in the window below:

1. Lens not pierced and not vice versa
2. The lens is plugged and not the other way around

3. Lens not pierced and vice versa 4. The lens is plugged and vice versa

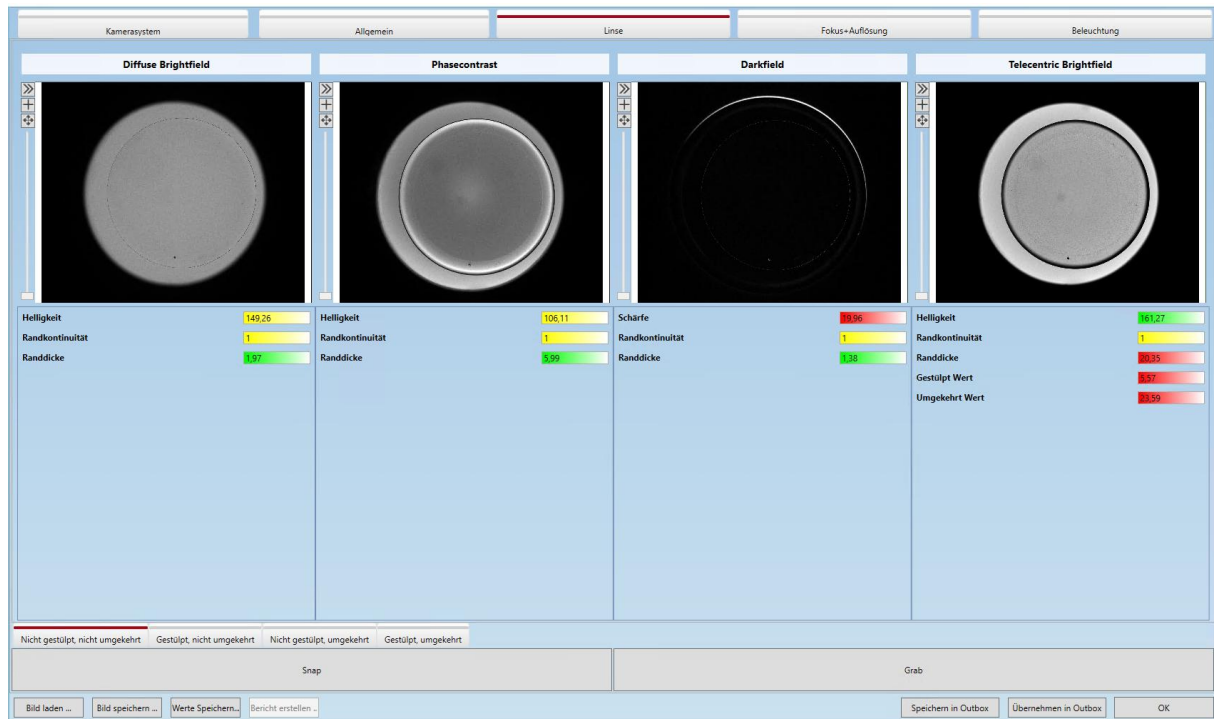


Figure 39: Table 'Linse' on the example of DSM Flex BV

4.2.4 Hardware jigs

The functionalities of the two Focus+Resolution tabs (Figure 41) and Lighting tabs (Figure 42) refer to recordings of a hardware jig. These two tabs are only available if the two files are present:

- "ImageSetup.conf" in the directory "..\Inbox\[machine name]\ApplicationSettings" and
- "ImageComparingTool.hdev" in the directory "..\LocalSettings\ImageComparing"

at the start of the software.

The behaviour and configuration of the limit values for the calculations are analogous to the tab lens (see 4.2.3.). In addition, it is possible in the ImageSetup.conf to hide the displayed values on the basis of the currently logged-in user level. Figure 40 shows an example of an output that is only displayed on the GUI if the currently logged in user is at least part of the administrator group. This output is therefore invisible to service and user groups.

```

"Name": "Verdrehung des Kreuzes",
"HalconOutputParameter": "CrossHairTorsionValues",
"Values": [
  {
    "Image": "Telecentric Brightfield",
    "Min": 0.0,
    "OptimumMin": 0.0,
    "OptimumMax": 10.0,
    "Max": 25.0,
    "IsDisabled": false,
    "IsVisibleFromUserLevel": "Administrator"
  },
  {
    "Image": "Phasecontrast",
    "Min": 0.0,
    "OptimumMin": 0.0,
    "OptimumMax": 10.0,
    "Max": 25.0,
    "IsDisabled": false,
    "IsVisibleFromUserLevel": "Administrator"
  },
  {
    "Image": "Diffuse Brightfield",
    "Min": 0.0,
    "OptimumMin": 0.0,
    "OptimumMax": 10.0,
    "Max": 25.0,
    "IsDisabled": false,
    "IsVisibleFromUserLevel": "Administrator"
  }
]

```

Figure 40: Example of an output that is only visible from the administrator level

Depending on the button group, the snap and grab buttons take either three images or one image.

With image loading or image saving, all four images of the surface are always loaded or saved. The file name has the extension _FJ#1 or _FJ-D2#2. The last number continues to refer to the camera type:

- #1 = telecentric light field
- #2 = dark field
- #3 = diffuse light field
- #4 = phase contrast

A data package is created for each camera head via the button "Back up data package". The data package contains: a PDF file, a CSV file, the pictures of the hardware jigs, the PFS files of the cameras, the lighting controller setting and the registration file of the camera head. A dialog can be used to select the location and file name for the data package.

The PDF report is temporarily stored in the software folder and opened automatically. All issues are stored in the CSV and PDF files, regardless of the user level of the currently logged in user.

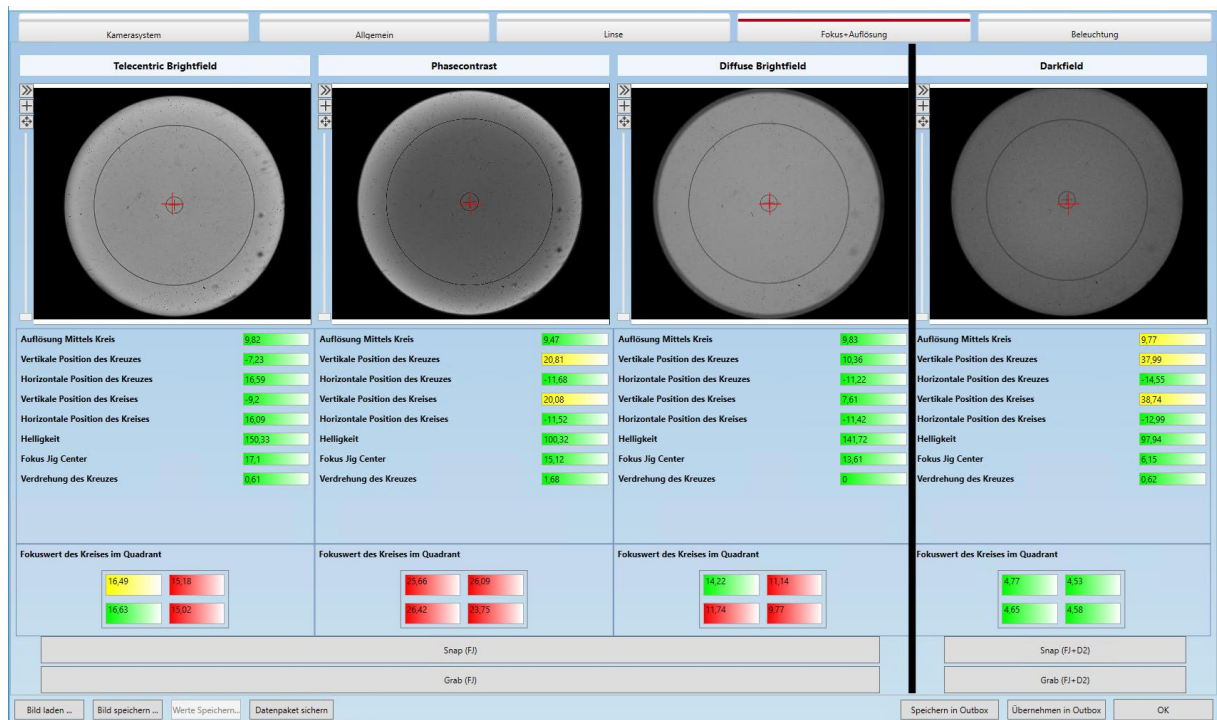


Figure 41: Tab Focus + Resolution

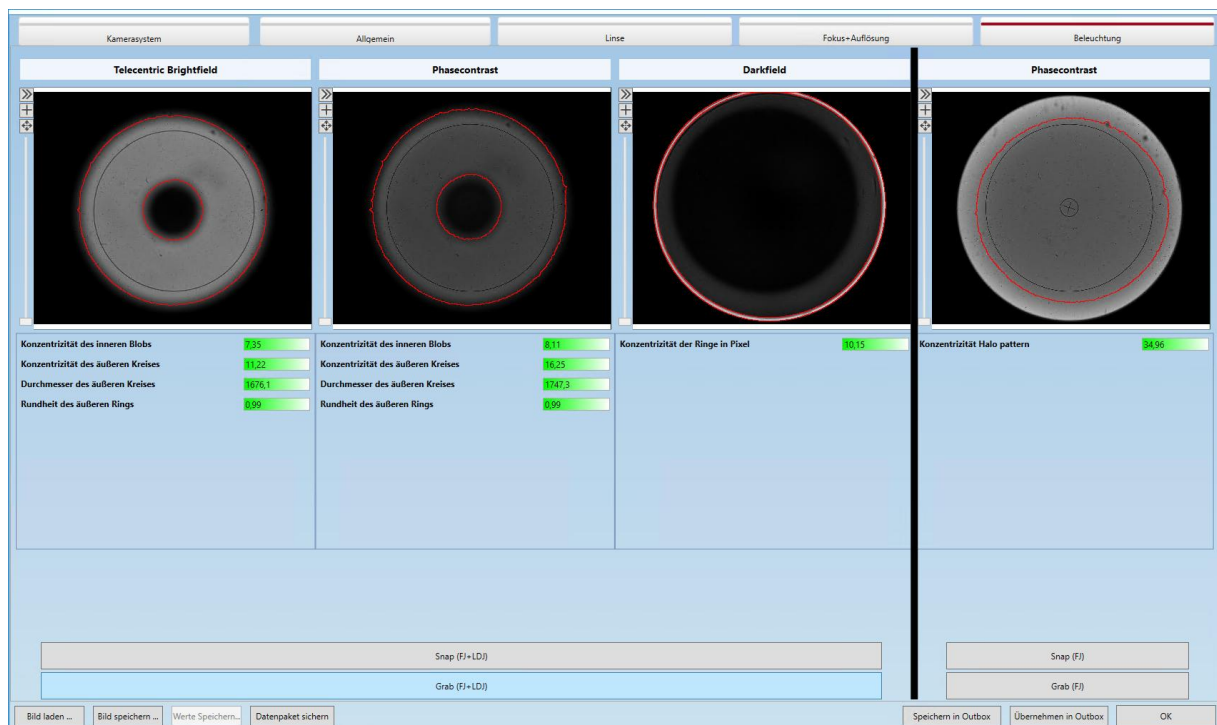


Figure 42: Lighting tab

4.3 General Settings

In the Settings window, general settings for the software are displayed and can also be changed there. For an overview of the settings see General System Settings, page 39.

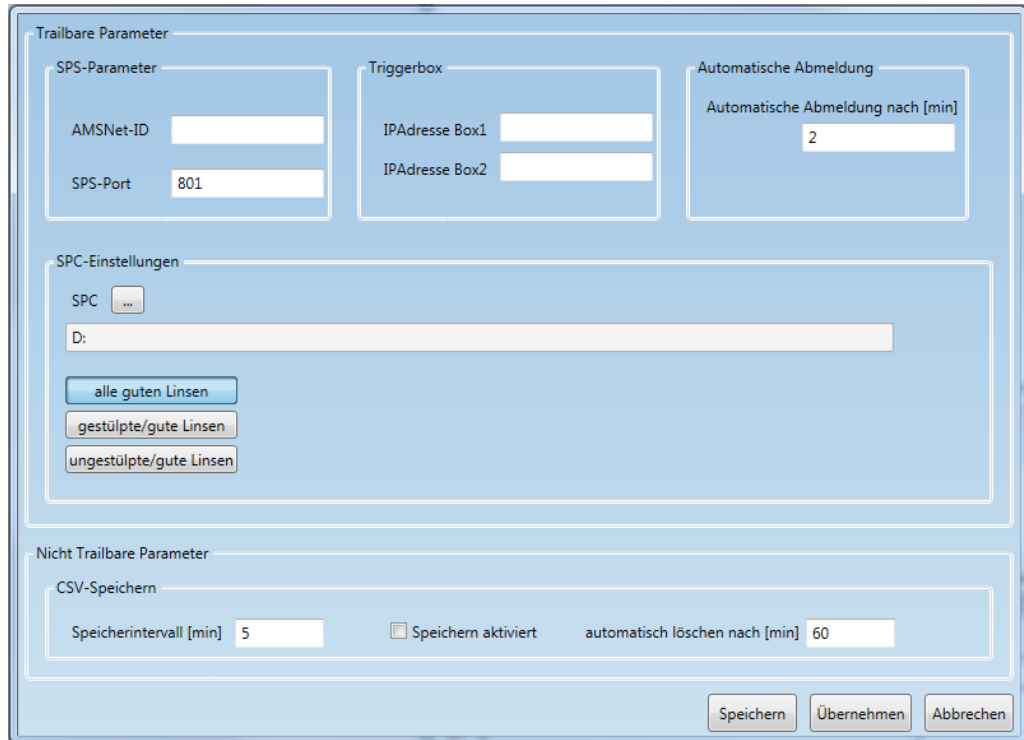


Figure 43: General settings dialog on the example of LSFlex BV

Depending on the system configuration, certain display elements of the window vary, such as the number of edit windows for the IP addresses of the LED controllers.

4.4 BV test

This function is used to perform a test of image processing scripts. When the button "BV Test" is activated, a window opens, which can vary depending on the system configuration. The following figure shows this for a camera head with four cameras:

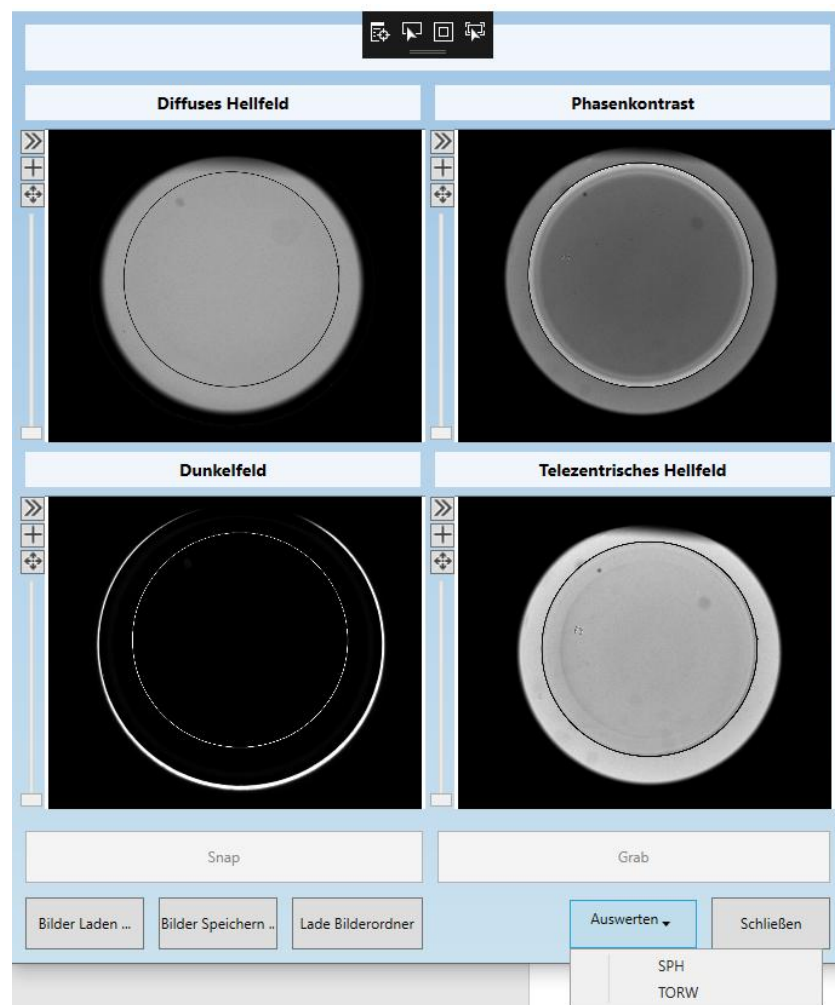


Figure 44: BV test

Using the snap button, images can be taken via the hardware of the system for use for the test. Alternatively, the button "Images Loading..." offers the possibility to load previously saved images from the hard disk. Using the Evaluate drop-down menu, the script to use can be selected. After the selection, the evaluation for the selected images is initiated. Subsequently, the results of the evaluation, as in normal production operations, are shown on the surface. A picture folder with more than one lens can be evaluated via the button "Load Image Folder...". After selecting the lens type, the evaluation starts automatically. The test can be stopped during execution (Figure 45). If the number of lenses in the image folder does not correspond to an integer multiple of the number of lenses per WT, the remaining positions are filled with "No test job".

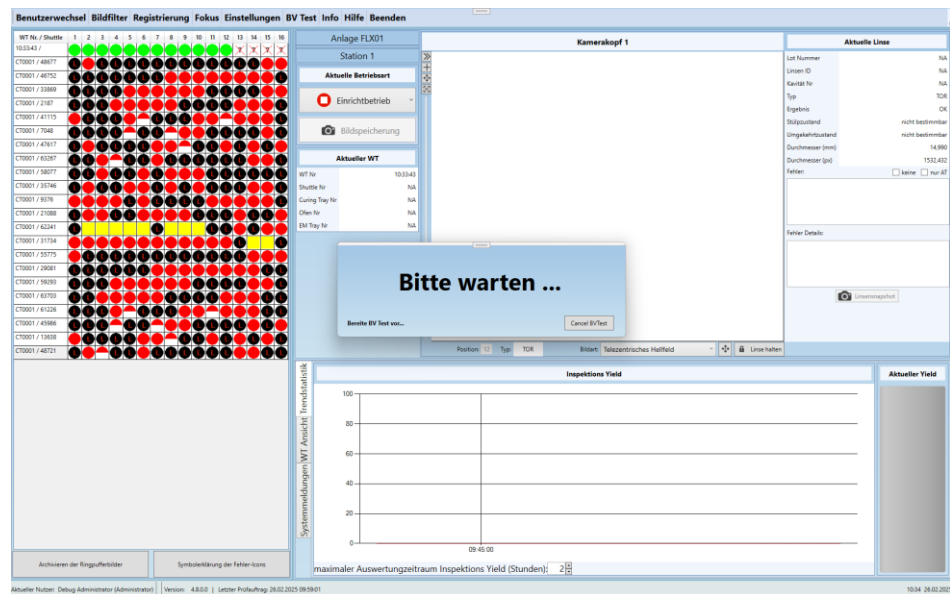


Figure 45: Evaluation of image folders at BV-Test

Live images can also be recorded via the grab button. While the grabbing is active, no lenses can be evaluated.

Note: Here the Halcon scripts are used, which were loaded into the automatic mode at the last change. A lens type is only available in the Evaluate drop-down menu if a Halcon script could be loaded for it.

4.5 Variable parameters

The following machine parameters can be changed in the software:

4.5.1 Camera Settings

Target values for the cameras are not explicitly listed. There are parameter files of the cameras that contain the default settings of the cameras by Emage Vision and have been set by CV.

Camera settings can be changed in the Camera System window by the user groups User, Service and Administrator. The following parameters are adjustable:

- Exposure time (28 - 10000000)
- Black level (0 - 600)
- Gain (0 – 490)
- Height (1 – (2058 - OffsetY))
- Width (1 – (2456 - OffsetX))
- OffsetX (0 – (2456 - width))
- OffsetY (0 – (2058 - altitude))
- Line Debouncer Time (0 - 100)

The OffsetX/OffsetY settings are directly related to height/width. So if the full height and width has been selected, OffsetX and OffsetY can only be 0. Conversely, the height and width cannot be maximum if OffsetX and OffsetY are not equal to 0.

The camera settings are stored in PFS files in the outbox.

Note: The values "height" and "width" are updated only with the next switch to automatic operation in the cameras.

4.5.2 Triggerbox Settings

Triggerbox settings can be changed in the Camera System window by the user groups User, Service and Administrator. The following parameters are adjustable:

- Pulse duration (pulse width) [µs] (min. 0 – max. 200000)
- Current used [A] (min. 0 – max. 4 A)

The trigger box settings are stored in the settings.conf file in the outbox.

4.5.3 General System Settings

General system settings can be changed by the Service and Administrator user groups in the Settings window. The following parameters are adjustable:

- IP address(s) of the trigger box(s) (xxx.xxx.xxx)
- AMS-Net ID of the PLC (xxx.xxx.xxx.x.x.x)
- Port of PLC (801, 811)
- Autologoff to N min (1 – 120)
- Path to save SPC images
- Memory interval [min] for CSV statistics (1 – 1440)
- Operating state of the CSV memory
- Minutes after which the CSV statistics should be deleted

The general system settings are stored in the Settings.conf file in the Outbox.

4.5.4 Image Filter Settings

Image filter settings can be changed by all user groups in the Image Filter window. The following parameters are adjustable:

- Position selection
- Selection of evaluation results
- Path to save the optimization images
- Number of optimization images to store
- Image Storage Mode
- Advanced image storage settings
 - Activation of extended image storage
 - Start date and start time of recurring image storage
 - Activation and time of interval storage
 - Model series
 - Daily: Number of days between two storages
 - Weekly: Number of weeks between two storages and the days of the week on which the storage should be active
 - Final criterion
 - Abort after certain amount of image storage
 - Abort on certain date

The image filter settings are stored in the localSettings.conf file in the LocalSettings folder.

4.5.5 Image Storage Settings

The image storage settings are stored in the ApplicationSettings.conf file during installation. The images can be saved either in the file format BMP or TIF. To do this, the value "ImageFilenameEnding/Ending" has to be set correctly for each camera and recipe. In the first camera, for example, the value "#1.bmp" or "#1.tif" is entered. It is important that the settings for all cameras are in the same format, so BMP and TIF are not mixed in a file ApplicationSettings.conf.

Additional metadata can only be stored in TIF images, but not in BMP images. Whether additional metadata has actually been saved can easily be checked in the properties (right-click). There the metadata are stored under the corresponding display field (see Figure 46). The metadata is stored in both automatic and set-up mode.

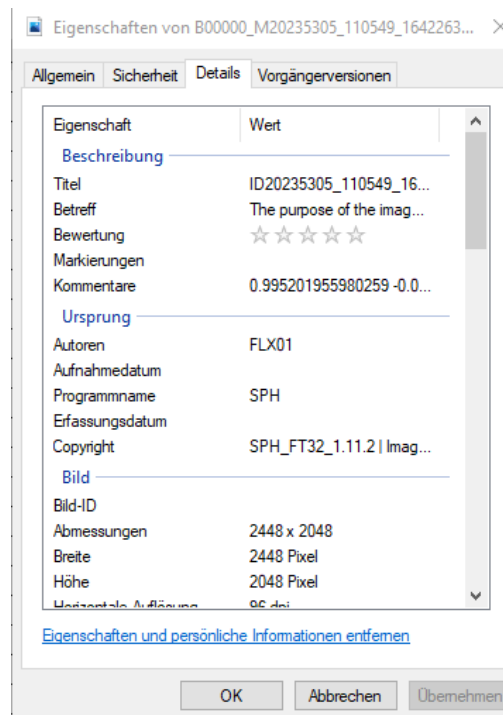


Figure 46: Properties of an image file

In the set-up operation, only the production plant, the time stamp of the image capture and the registration data is written to the metadata. The registration data is loaded directly from the inbox. For the other metadata, placeholders are used.

Registration data is loaded from the images in the evaluation tool. If no registration data or only incorrect ones are present, an error message is issued and the registration from the text file is used.

Note: When using TIF images, a registration file in the Inbox folder does not necessarily have to be present in evaluation mode, since for each single image it is attempted to extract the registration data from the metadata of the image. However, a warning message gives a hint in the system messages about the missing registration file in the Inbox folder. When using BMP images, a registration file must always be present.

4.5.6 Trigger timeout settings

The trigger timeout settings are stored in the ApplicationSettings.conf file. The following parameters are adjustable:

- CameraTriggerpulseDistanceMS: Time in milliseconds between the trigger pulses (as specified by the speed of the workpiece carriers).
- ImageProcessingTimeoutMS: Time in milliseconds in which the result of the evaluation is expected at the latest.

The 'total' timeout is calculated for each lens depending on the index:

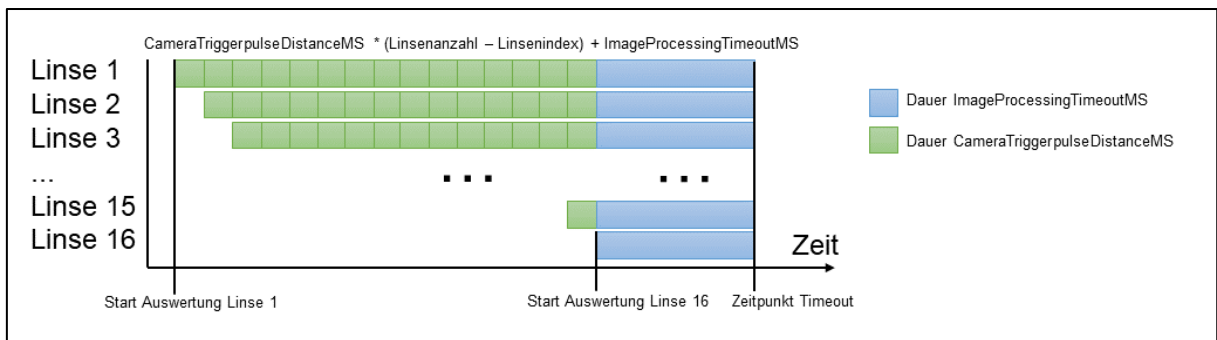


Figure 47: Illustration Timeout with 16 lenses

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